MILITARY SPECIFICATION

MINE CLEARANCE SYSTEM KIT (MCSK): FOR LANDING VEHICLE, TRACKED, PERSONNEL, MODEL 7A1 (LVTP7A1)

This specification is approved for use by U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the LVTP7A1 Mine Clearance System Kit (MCSK), exclusive of line charge containers, rocket motors, and ordnance material.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-130	-	Identification Marking of U.S. Military property
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-794	-	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	-	Environmental Test Method
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment
MS 52129	-	Connector, Receptacle, Electrical Intervehicle Power Cable

Beneficial comments (recommendationst additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps (LMA-1), Headquarter U.S. Marine Corps, Washington, D.C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Sea Systems Command

5417700 - Mine Clearance System Kit Installation, LVTP7A1 with Associated Data List

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directd by the contracting officer) ,

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. The contractor shall furnish two sample units for first article inspection and approval (see 4.5 and 6.3). The first articles shall be manufactured using the same methods, procedures, and tooling proposed for production. Approval of first articles by the procuring activity shall not relieve the contractor of the obligation to supply equipment conforming to this specification. Changes or deviations in production units from the approved first articles shall require written authorization by the procuring activity. No rockets shall be launched or line charges detonated for first article approval.

3.2 Materials. Except as specified herein, materials shall be selected by the manufacturer. Materials selected shall be capable of meeting the performance requirements specified herein.

3.3 Design and construction. The MCSK shall be fabricated and assembled in accordance with Drawing No. 5417700 and shall consist of the following major components:

(a) Launcher housing
(b) Rocket launcher assembly
(c) Hydraulic power supply
(d) Electrical control system
(e) Support equipment

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3.3.1 Launcher housing. The launcher housing assembly shall be equipped with two hatches which automatically open or close when the rocket launcher assembly is hydraulically raised or lowered. The hatches must be manually unlatched before deployment begins. The housing shall be equipped with a gasket which provides watertight protection when mated to the LVTP7A1 cargo hatch opening.

3.3.2 Rocket launcher assembly. The rocket launcher shall consist of two support arms and a launcher platform upon which three rocket motors ar loaded. Each rocket motor shall project a line charge when fired. The rocket launcher assembly shall contain two hydraulic cylinders. One cylinder shall be used to raise/lower the support arms, along with the hatches, while the other shall be used to angle the launcher platform into a safe to launch pcsition-

3.3.3 Hydraulic power supply. Hydraulic power to raise and lower the launcher shall be provided by a reversible motor pump which is driven off the vehicle electrical system. A hand pump shall provide a mechanical backup method of raising and lowering the launcher.

3.3.4 Electrical control system. Electrical power (nominal 24 direct-current volts (Vdc) for the MCSK shall be obtained from the vehicle electrical system, through the aft slave receptacle. A NATO standard plug, conforming to MS52129-1, is required to interface with the vehicle system. The electrical control system shall include a launcher control box and an arm switch.

3.3.4.1 Launcher control box. The control box shall be located at the operator’s station to control launcher movement, to initiate rocket launch, and to detonate the deployed line charge. Also test circuits shall be provided for testing the electrical continuity of the system. The launcher control box shall provide the following controls and indicators.

3.3.4.1.1 Power switch. The power switch shall be a two-position toggle switch. When the switch is placed in the ON position, system power shall be applied and the green indicator light shall be on. When the switch is placed in the OFF position, system power shall be removed. This switch shall also be a circuit breaker and shall "open" should there be an excessive current demand from the system.

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3.3.4.1.2 Launcher control switch. When this switch is held in RAISE position, the hydraulic power supply shall operate and the launcher shall rise. When desired launcher angle is obtained, the switch shall be released and return to the neutral (OFF) position automatically. Pushing downward on the switch shall lower the launcher.

3.3.4.1.3 Support Indicator. The support up indicator (green light) shall be illuminated when the support is in the fully raised position.

3.3.4.1.4 Angle OK indicator. The angle OK indicator (green light) shall be illuminated when the launcher is in a safe-to-launch elevation between 42 degrees to 63 degrees with respect to the horizontal.

3.3.4.1.5 Rocket elevation meter. The rocket elevation meter shall indicate the angle of the launcher with respect to the horizontal. STOW and RAISE limits shall be marked on the dial face to provide guidance as to which direction to move the control to obtain the desired angle. The LAUNCH section on the dial shall be colored green, luminous, and shall cover the sector of 42 degrees to 63 degrees. The short vertical line on the dial face within the LAUNCH section shall indicate the optimum angle of launch of 47 degrees.

3.3.4.1.6 Line charge/rocket select switch. The line charge/ rocket select switch shall be pull and turn, four-position rotary type switch. It shall be manually operated to apply power to one of the three rocket firing and line charge detonating circuits.

3.3.4.1.7 Rocket-armed indicator. The rocket armed indicator shall show an amber light when the arm switch is held in the ROCKET position.

3.3.4.1.8 Launch rocket switch. The launch rocket switch shall be a spring-return push button switch. To launch a rocket, the arm switch must be held in the ROCKET position and then the launch rocket switch must be pushed.

3.3.4.1.9 Line charge armed indicator. The line charge armed indicator shall show an amber light when the arm switch is held in the LINE CHARGE position.

3.3.4.1.10 Line charge detonate switch. The detonate line. charge switch shall be a spring-return push button switch. To detonate a line charge, the arm switch myst be in the LINE CHARGE position and then the detonate line charge switch must be pushed.

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3.3.4.1.11 Panel lights. Panel lights shall be located at the upper right and left hand corners of the control box. The lights shall be turned on by pulling out the protective shields over the lights. The lights shall be turned off by pushing in the shields.

3.3.4.2 Arm switch. The arm switch shall be a three-position, center-off, spring-return toggle switch and shall be mounted on the vehicle hull to permit arming of either the rocket launch or line charge detonation circuits. The switch shall be covered by a hinged safety guard which must be swung upward to allow switch operation. The switch must be held in the forward (LINE CHARGE ARMED) position OE rearward (ROCKET ARMED) position and when released shall snap back to the center off position. When the switch is held in the ROCKET ARMED position, the rocket armed light shall be illuminated, indicating the rocket is ready to be launched. When the switch is held in the LINE CHARGE ARMED position, the line charge armed light shall be illuminated, indicating the line charge is ready to be detonated. The arm switch shall be installed by means of a keyed electrical connector “which prevents incorrect installation.

3.3.5 Support equipment. MCSK support equipment shall consist of a pallet rail system, stowable loading ramps, fore and aft guards, cables and turnbuckle assemblies; and a capstan. This equipment shall be used in the installation and securing of the line charge containers and for preventing damage to the launcher housing during deployment.

3.3.6 Interchangeability. All parts, components, and assemblies having the same apart number shall be functionally and physically watertight requirements.

3.4 Performance. The MCSK shall meet the following performance requirements.

3.4.1 Watertight integrity. The MCSK shall meet the following watertight requirements.

3.4.1.1 Launcher housing/vehicle hull interface. The seal between the launcher housing and the hull of the LVTP7A1 shall pass no water when tested in accordance with 4.5.4.1.

3.4.1.2 Launcher hatches. The seal around the launcher hatches shall pass no water when tested in accordance with 4.5.4.1.

3.4.2 Capstan/container loading. The time required to load a fully loaded line charge container Up the ramp and through the LCTP7A1 ramp opening and into the forward most position shall not exceed 36 seconds when using the capstan motor.

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3.4.3 Capstan/container unloading. The time required to unload the equipment of a fully loaded line charge container through the LVTP7A1 ramp opening shall not exceed 36 seconds when using the capstan motor.

3.4.4 Launcher raising (powered). The time required for the loaded laucher (three rockers on the rails) to open the unlatched launcher doors and continue raising until it contacts the launcher stop at the aft end of the launcher housing shall not exceed 33 seconds, when driven by the motor driven hydraulic pump.

3.4.5 Launcher lowering(powered). The time required for the load launcher to lower to its maximum downward position and close the launcher doors shall not exceed 33 seconds when driven by the motor driven hydraulic pump.

3.4.6 launcher raising (manual). Not more than 175 pump strokes shall be required to raise the loaded launcher to its full up position when using the emergency hand pump.

3.4.7 Hydraulic motor pump The hydraulic motor pump shall provide a flow of 1.8 gallons per minute (gpm) ± 0.1 gpm at 2000 ±50 pound-force per square inch gage (psig) when supplied with a nominal 24 Vdc.

3.4.8 Hydraulic hand pump. The hydraulic hand pump shall provide a minimum flow of 1.25 cubic inches per cycle (2 strokes) at 1500 psig.

3.4.9 Hydraulic fluid loss. Hydraulic fluid loss shall not exceed 0.05 cc (1 drop) for a single powered raise/lower cycle and a single manual raise/lower cycle.

3.4.10 Environmental characteristics. The MCSK shall withstand the following operating and storage environmental conditions.

3.4.10.1 Operating. The MCSK shall be capable of performing all of the requirements specified herein while exposed to the following environnental conditions:

(a) Temperature: -54°C (-65°F) to +71°C (+160°F)
(b) Humidity: Up to 95 percent with a temperature of 60°C (140°F), including condensation due to temperature changes
(c) Vibration: Levels and duration as specified in MIL-STD-810, method 514.2, Procedure VIII.
(d) Shock: Levels and duration as specified in MIL-STD-810, Method 516.2, Procedure II.
(e) Salt fog: Levels as specified in MIL-STD-810.

3.4.10.2 Storage. The MCSK shall be able to withstand storage in the following environmental conditions and be able to perform as specified herein after such exposure:

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(a) Temperature: -62°C (-80°F) to +71°C (+160°F)
(b) Humidity: Same as 3.4.10.1
(c) Vibration: Same as 3.4.10.1
(d) Shock: Same as 3.4.10.1
(e) Salt fog: Same as 3.4.10.1

3.5 Weight. The MCSK shall not weigh more than 10,000 pounds.

3.6 Marking. MCSK equipment shall be marked in accordance with the requirements of MIL-STD-180.

3.7 Government-loaned property. The following equipment shall be loaned by the Government for first article and quality conformance tests :

Item	Quantity/Unit
M69 Linear Demolition Charge (inert) and Container	3 each
MK 22 Mod 2 (certified inert by the Government and ballasted to identical weight of a live rocket)	3 each
LVTP7A1 Landing Vehicle, Tracked, Personnel	1 each

3.8 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may *use* his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article (4.3)
(b) Quality conformance (4.4)

4.3 First article tests. First article tests shall be performed by the contractor, after the award of contract and prior to production, at a location acceptable to the Government. First article approval is valid only in the contract under which it

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is granted, unless extended by the Government to other contracts. First article tests shall be performed on sample units which have been produced with equipment and procedures normally used in production.

4.3.1 Nonconformance of first articles. In the event the first articles fail to meet the requirements of 4.3.2, the contractor shall submit plans for corrective action to the procuring activity for approval. Any failure in the inspections and tests shall be cause for the unit to be rejected. Any rejected unit may be reworked and resubmitted for first article testing. Production testing shall not comence until first articles have passed first article inspection and tests.

4.3.2 First article inspection and testing. The first articles shall be subjected to the examination and tests specified in Table I, in the order shown. Each first article shall be installed and checked out on the LVTP7A1 vehicle prior to first article test. All first article tests shall be conducted on each first article to determine conformance with the requirements specified herein.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in Table I. Each MCSK accepted by the Government shall have successfully completed all quality conformance inspections and tests.

4.4.1 Rejection and retest. Any MCSK which fails any of the qual conformance tests shall be rejected. The rejected MCSK may be reworked or have parts replaced to correct defects. The MCSK shall be completely retested to demonstrate that the requirement meets the requirements specified herein.

4.5 Inspection and test

4.5.1 Inspection and test conditions. Unless otherwise specified herein, all inspections and tests shall be performed in accordance with the test conditions specified in MIL-S.TD-810.

4.5.2 Visual examination. The system shall be visually examined to verify conformance to 3.3 and 3.7.

4.5.3 Material inspection. The contractor shall certify that all materials and parts used in the fabrication of the MCSK are in accordance with requirements specified herein. Material or part manufacturer's certification of test data will be acceptable for fulfilling this requirement. Acceptance or approval of any material or part shall not be construed as a guarantee of acceptance of the MCSK.

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TABLE I. Inspection and tests

Subject	Requirement (Para)	Test (Para)	First Article	Quality Conformance
Visual examinations				
Design and construction	3.3	4.5.2	X	X
Material inspection	3.3	4.5.3	X	X
Workmanship	3.8	4.5.2	X	X
Equipment tests				
Watertight integrity	3.4.1	4.5.4.1	X	X
Capstan/container loading	3.4.2	4.5.4.2	X (10 cycles)	
Capstan/container unloading	3.4.3	4.5.4.3	X	
Launcher raising (powered)	3.4.4	4.5.4.4	X (25 cycles)	X (5 cycles
Launcher lowering (powered)	3.4.5	4.5.4.5	X	X
Launcher raising (manual)	3.4.6	4.5.4.6	X	X
Hydraulic motor pump	3.4.7	4.5.4.7	X	X
Hydraulic hand pump	3.4.8	4.5.4.8	X (10 cycles)	X (1 cycle)
Hydraulic fluid loss	3.4.9	4.5.4.9	X	X
Power switch	3.3.4.1.1	4.5.4.10	X	X
Launcher control switch	3.3.4.1.2	4.5.4.11	X	X
Support up indicator	3.3.4.1.3	4.5.4.12	X	X
Angle OK indicator	3.3.4.1.4	4.5.4.13	X	X
Rocket elecation meter	3.3.4.1.5	4.5.4.14	X	X
Line charge/rocket select switch	3.3.4.1.6	4.5.4.15	X	X
Rocket-armed indicator	3.3.4.1.7	4.5.4.16	X	X
Rocket launch switch	3.3.4.1.8	4.5.4.17	X	X
Line charge armed indicator	3.3.4.1.9	4.5.4.18	X	X
Line charge detonate switch	3.3.4.1.10	4.5.4.19	X	X
Panel lights	3.3.4.1.11	4.5.4.20	X	X
Arm switch	3.3.4.2	4.5.4.21	X	X
Environmental Tests	3.4.10	4.5.4.22	X	

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4.5.4 Tests.

4.5.4.1 Watertight integrity. A 200-gpm water source (±20 gpm) discharged through a hose with a 1-inch nozzle shall be directed at the seal between the launcher housing and the vehicle hull (launcher hatches to be closed and latched shut). The stream shall be directed for 1 minute for every 5 feet of seam with the nozzle at a distance of 3 feet from the seam at various angles. The same procedure shall be repeated for the seal around the Launcher hatches. Leakage shall be in accordance with requirements of 3.4.1.

4.5.4.2 Capstan/container loading test. The time-required to load the Government-loaned M69 Linear Demolition Charge (inert) container through the LVTP7A1 ramp opening shall be determined using Test Method No. 1 in the Appendix.

4.5.4.3 Capstan/container unloading test The elapsed time required to unload the Government-loaned container through the ramp opening shall be determined using Test Method No. 2 in the Appendix.

4.5.4.4 Powered launcher raising test. The elapsed time required to raise the launcher while loaded. with three Government-loaned inert rockets shall be determined using Test Method No. 4 in the Appendix. The hatches shall lock in the vertical position as the launcher is raised with all cams and rollers functioning smoothly.

4.5.4.5 Powered launcher lowering test. The elapsed time required to lower the launcher while loaded with three Government loaned inert rockets shall be determined using Test Method No. 4 in the Appendix. The hatches shall release from their verticle locked position and lower with the launcher unit. All cams and rollers shall function smoothly.

4.5.4.6 Launcher raising manual test. Performance of the launcher emergency hand pump shall be verified using Test Method No. 3 in the Appendix, and shall be in accordance with 3.4.6.

4.5.4.7 Hydraulic motor pump test. The hydraulic motor pump shall be tested to verify compliance with requirements of 3.4.6

4.5.4.8 Hydraulic hand pump test. The hydraulic hand pump shall be tested to verify compliance with requirements of 3.4.7.

4.5.4.9 Hydraulic fluid leakage test. The MCSK shall be tested to verify compliance with fluid loss requirements of 3.4.8.

4.5.4.10 Power switch test. Performance of the power switch shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.1.

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4.5.4.11 Launcher control switch test. Performance of the launcher control switch shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.2.

4.5.4.12 Support up indicator test. Performance of the support up indicator shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.3.

4.5.4.13 Angle OK indicator test. Performance of the angle. OK indicator shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.4.

4.5.4.14 Rocket elevation meter test. Performance of the rocket elevation meter shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.5.

4.5.4.15 Line charge/rocket select switch test. Performance of the line charge/rocket select switch shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.6.

4.5.4.16 Rocket-armed indicator test. Performance of the rocket-armed indicator shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.7.

4.5.4.7 Rocket launch switch test. Performance of the rocket launch switch shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.8.

4.5.4.18 Line charge armed indicator test. Performance of the line charge armed indicator shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.9.

4.5.4.19 Line charge detonate switch test. Performance of the line charge detonate switch shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.10.

4.5.4.20 Panel lights test. Performance of the panel lights shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.1.11.

4.5.4.21 Arm switch test. Performance of the arm switch shall be verified using Test Method No. 4 in the Appendix. Performance shall be in accordance with 3.3.4.2.

4.5.4.22 Environmental tests.

(a) Temperature. The nonoperating and operating temperature tests shall be in accordance with method 501.1 and method 502.1 of MIL-STD-810.

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(b) Humidity. The humidity test shall be in accordance with method 507.1, Procedure IV of MIL-STD -810.

(c) Vibration. The vibration test shall be in accordance with method 514.2, Procedure VIII, Sinusoidal cycling, Test time schedule A, Test Curve W, of MIL-STD-810.

(d) Shock. The shock test shall be in accordance with method 516.2, Procedure III of MIL-STD-810.

(e) Salt fog. The salt fog test specified in method 509.1 of MIL-STD-810 shall be applied to the finishes and coatings on parts of frames and enclosed structures as finally assembled for use.

4.5.5 Failure. A failure is defined as any deviation from the performance characteristics specified in 3.4 or any variation from the testing requirements as identified in 4.3 and 4.4 or physical breakdown or wear of any part of the equipment.

5. PACKAGING

5.1 Preservation-packaging. Preservation and packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The complete MCSK unit shall be-preserved and packaged in accordance with the level A requirements of MIL-STD-794.

5.1.2 Commercial. The complete MCSK shall be preserved and packaged in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A or commercial (see 6.2).

5.2.1 Level A. The complete MCSK, preserved and packaged as specified in 5.1.1, shall be packed to conform to the level A requirements of MIL-STD-794.

5.2.2 Commercial. The complete MCSK, packages as specified in 5.1.2 shall be packed in conformance with MIL-STD-1188.

5.3 Marking. Marking for pckaging and packing shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The MCSK described herein will enable the LVTP7A1 vehicle to be converted quickly from a personnel carrier to one having the capability of clearing minefield and other obstructions emplaced on the beach and inland areas.

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6.2 Ordering data.

6. 2.1 Acquisition requirements. Acquisition documents should specify the following.

a. Title, number, and date of this specification.

b. Level of preservation, packaging, and packing [5.1 and 5.2)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Option
3.1	First article Inspection Procedure	DI-T-4901	----
3.1	First article Inspection Report	DI-T-4902	----
4.5	Test plan	DI-T-5204	----
4.5	Test report	DI-T-5247	----

MIL-M-29399(MC)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. Any changes or deviations of production MCSK's from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the supplier of his obligation to furnish MCSK'S conforming to this specification.

Preparing activity:
Navy - MC
project No. 1055-NO65

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APPENDIX

MINE CLEARANCE SYSTEM KIT (MCSK): FOR LANDING VEHICLE, TRACKED, PERSONNEL, MODEL 7A1 (LVTP7A1)

Scope: The test methods contained in this appendix are to determine whether MCSK'S procured under this specification conform to the requirements specified herein.

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TEST METHOD NO. I

LOADING LINE CHARGE CONTAINERS THROUGH THE RAMP OPENING INTO LVTP7A1

TEST PROCEDURE:

a. Lower rear ramp of vehicle. Make sure the troop seats (on port and starboard sides) are lowered.

b. Unfasten straps securing loading ramps and remove ramps from stowage brackets.

c. Insert loading ramp ends into the support rail sockets.

d. Secure ramp connector between the two loading ramps.

e. Turn CAPSTAN switch on momentarily to assure capstan is operable.

f. Place capstan rope hook at pallet to be loaded and fairlead rope from container to capstan.

g. Align container skids with loading ramps. (Check to assure that forward end of container is facing into the vehicle ramp opening .)

h. Attach the capstan rope hook to the forward skid ring of the container.

i. Take up the slack in the capstan rope and pass free end under and around the capstan head. Take at least three turns counterclockwise around the capstan head.

j. Turn on capstan power. Apply tension to free end of rope, and begin pulling container onto loading ramps. Continue pulling the container up the ramps and onto the support rails. (Rate of movement of the container and rope tension can be manually controlled by slacking the capstan rope, allowing rope to slip on capstan.)

k. Stop the container at a position just inside the rear ramp opening. (Container may be stopped by slacking rope or by turning off capstan power.)

l. Disengage the hook from the forward skid ring of container. Pass hook under container and reattach to the aft skid ring.

m. slowly take up slack in rope (with capstan) and proceed to move container into desired position. The forward container

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must be positioned so that the skids butt against the stops at the forward end of the support rails. When the container being loaded is in position, release tension on rope and turn off the capstan power. Disengage rope from capstan and container.

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TEST METHOD NO. 2

OFF LOADING LINE CHARGE CONTAINERS

Note: Off-loading of containers containing- packed line charges through the ramp opening requires a different procedure than off-loading empty containers. Empty containers can be pulled from the vehicle by the use of suitable line (or capstan rope. off-loading empty containers can be accomplished manually or with a power source. The following procedures apply when using the capstan to off-load packed charges.

TEST PROCEDURE:

a. Lower ramp and position loading ramps as specified in Test Method No. 1 a through e.

b. With suitable available manilla line or other strapping, secure the loading ramps to the vehicle ramp at a point near the lowered end of the vehicle ramp (to prevent the loading ramps from slipping upward during off-loading of packed containers) .

c. Secure the snatch block to the ramp connector.

d. Attach the hook on the capstan rope to the aft skid ring of the aft container.

e Fairlead the free end of the rope through the pulley block, forward under the containers and to the capstan. Take at least three turns around the capstan head (counterclockwise).

f. Turn on capstan power. Apply tension to free end of rope and pull the container along the support rails until the skids contact the loading ramps. With the container at the ramps, proceed as follows:

(1) Pass the free ends of a suitable safety rope (length approximately 50 feet; approximate diameter 3/4-inch nylon) through the lifting rings of the container (one end through each ring).

(2) place the loop of the rope (formed between the two lifting rings) across the aft side of the container. Take up slack.

(3) Pass the free ends of the rope over the vehicle cleats at the stern of the vehicle), port side to port cleat and star-board side to starboard cleat.

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(4) Station a line handler on each side of the ramp to handle safety line during off-loading.

g. With the two line handlers manning the safety rope, continue to pull the container out of the vehicle onto the ramps (using capstan). Continue off-loading until container is near end of ramps.

h. Disengage capstan rope and safety rope from container.

i. Repeat actions described in c. through g., of this test, as required to off-load remaining containers.

j. Turn off capstan power. Remove and stow loading ramps and close ramp.

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TEST METHOD NO.3

EMERGENCY HAND PUMP FUNCTIONAL TEST

TEST PROCEDURES:

a. Alert crew that launcher is going to be raised.

b. Set the hand pump manual four-way valve to the RAISE position. Operate the manual hand pump to elevate the launcher The launcher support shall raise opening the launcher doors to detent position and continue raising until it contacts its stop at the aft end of the launcher housing. Continue operation of the manual pump to rotate the launcher rack until the mechancal pendulum pointer is positioned over the scribed line on the pendulum dial.

c. Alert crew that launcher is going to be lowered.

d. Rotate the manual four-way valve handle to the LOWER position and lower the launcher to the stowed position by operating the hand pump.The launcher shall rotate to its maximum downward position and then the launcher support will lower to its stowed position within the launcher housing. During lowering, cams on the support will engage the launcher door cam followers, release the door detents and close the doors. Turn the manual four-way value to the OFF position.

e. Make certain the hand pump manual control valve located above the hydraulic power supply for the kits is in the OFF posi-tion.

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TEST METHOD NO.4

SYSTEM FUNCTIONAL CHECK OUT PROCEDURES

TEST PROCEDURES:

a. start the vehicle engine and allow it to warm up. Set engine at approximately 1800 rpm.

b. Press the red SLAVE OUTPUT POWER button on the distribution box.

c. press the red circuit breaker reset button on the distribution box at the capstan.

d. At the launcher control box, set the POWER switch in the ON position.

(1) The green POWER indicator shall illuminate and the two panel illuminating lights shall light when their shields are pulled out.

(2) Press each panel indicator light to make certain the lamps are operational.

(3) Actuate the switch on the rear of the utility light located at the aft port corner of the launcher housing to verify that light is operative.

e. Alert crew that launcher is going to be raised.

f. With the launcher doors unlatched and both door cam followers engaged in the launcher cams, hold the launcher switch in the RAISE position.

(1) The launcher support shall raise opening the launcher doors to the detent position and continue raising until it contacts its stop at the aft end of the launcher housing.

(2) The green SUPPORT UP light on the launcher control box shall illuminate and the launcher shall rotate upwards.

(3) When the angle of the launcher is between 42 degrees and 63 degrees the ANGLE OK light shall be illuminated and the angle meter needle shall be on the green launch area of the meter scale. The mechanical pendulum indicator on the launcher shall be in the white (“safe to launch”) area of its scale.

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(4) When the angle of the launcher is at 47 degrees, the optimum launch angle, the angle meter needle shall be over the short vertical line above the word LAUNCH on the meter scale. The mechanical pendulum indicator shall be over the black scribed line on its scale.

(5) Normal time to raise is 33 seconds.

g. At the launcher control box, set the LINE CHARGE/ROCKET SELECT switch to the orange bar (forward line charge and port rocket).

(1) Hold the ARM switch on the vehicle hull at the left of the launcher control box at the ROCKET position. At the launcher control box, the ROCKET ARMED light shall be lit.

(2) Press the ROCKET LAUNCH button on the launcher control box.

(3) At the port rocket filter box, only that rocket test plug shall be lit.

h. At the launcher control box, release the ROCKET LAUNCH button.

(1) Hold the ARM switch at the LINE CHARGE position. At the launcher control box, the LINE CHARGE ARMED light shall be lit.

(2) Press the LINE CHARGE DETONATE button.

(3) Only the line charge test plug at the forward line charge receptacle shall be lit.

(4) Release both the LINE CHARGE DETONATE button and the ARM switch.

(5) At the launcher control box, set the LINE CHARGE/ ROCKET SELECT switch to the blue bar and repeat steps g and h for the center line charge and the starboard rocket.

(6) At the launcher control box, set the LINE CHARGE/ ROCKET SELECT switch to the gray bar and repeat steps g and h for the aft line charge and center rocket.

(7) Turn the LINE CHARGE/ROCKET SELECT switch to OFF.

i. At the launcher control box, hold the launcher switch to the RAISE position.

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(1) The launcher shall rotate upwards and the ANGLE OK light shall go out when the angle of the launcher with respect to the horizontal exceeds 63 degrees.

(2) Repeat step f. No test plugs for any rocket shall light. Line charge test plugs shall continue to light.

(3) At the launcher control box, hold the launcher switch to the LOWER position. The launcher shall rotate downwards. The ANGLE OK light shall come on when the launcher is at an angle of 63 degrees with the horizontal. It shall remain on until the angle with horizontal is less than 42 degrees at which time the light shall go out.

(4) With the launcher at an angle with the horizontal of less than 42 degrees (ANGLE OK light not lit), repeat step g. No test plugs for any rocket shall light. Line charge test plugs shall continue to light.

j. Alert crew that launcher is to be stowed. At the launcher control box, hold the launcher switch at the LOWER position. The launcher shall rotate to its maximum downward position and then the launcher support shall lower to its stowed position within the launcher housing. During lowering, cams on the support shall engage the launcher door cam followers, release the door detents and close the doors.

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